

We observe two independent samples: prior-grade scores $\{X_i, w_i\}$ and current-grade scores $\{Y_j, w'_j\}$ for the same subgroup (e.g., a TIMSS country), but *no student-level pairs* (X_i, Y_i) are available.

We map scores to reference-percentiles using fixed reference marginals, $U = F_X^{\text{ref}}(X)$ and $V = F_Y^{\text{ref}}(Y)$, so that distributional shifts remain visible across groups and periods.

The weighted empirical distributions of U (prior; blue) and V (current; red) are therefore the only observables; all subsequent inference proceeds from these marginals plus a baseline dependence template.